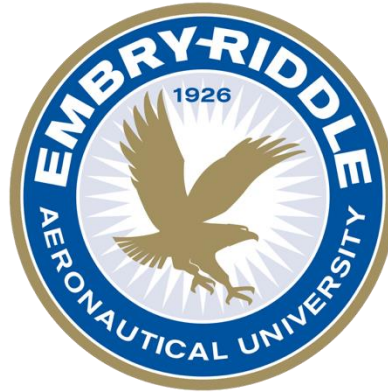
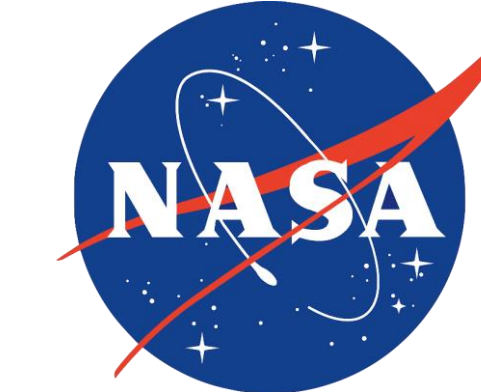




Characterizing the Solar Sources and Geospace Response of Isolated Stream Interaction Regions with Embedded Periodic Mesoscale Solar Wind Structures



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~ Overview ~

Stream Interaction Regions (SIRs) are compression regions from fast/slow solar wind interaction. They are less intense than CMEs but efficient drivers of recurrent geomagnetic activity. SIRs are embedded in a mesoscale-structured solar wind. One signature of that structure, periodic density structures (PDSs), potentially preconditions SIRs to be more geoeffective.

Main Research Objective: Characterize the solar wind sources upstream of SIRs and the associated magnetospheric responses, using ADAPT-WSA coronal modeling, solar wind observations, and magnetospheric indices.

Keywords: *Periodic Density Structures, Stream Interaction Regions, Coronal Source Region*

~ Data ~

1. SIR List from Christos and Simone's 2025 Paper²
2. One-Minute OMNI Data:
 - B, V, AE, AU, AL, Sym H, Alpha Proton Ratio, and Plasma Beta
 - Calculated total energy input into the magnetosphere via epsilon (solar wind coupling function)
3. One-Hour OMNI Data:
 - Magnetosphere parameters Kp (3-hour product) and Dst
4. Three Substorm Lists:
 - Newell, Ohtani, and Forsyth from SuperMAG
5. ADAPT-WSA Model Output Files
6. Source Region Parameters:
 - Expansion factor, Squashing factor, B footpoint, Coronal hole distance, Current sheet distance

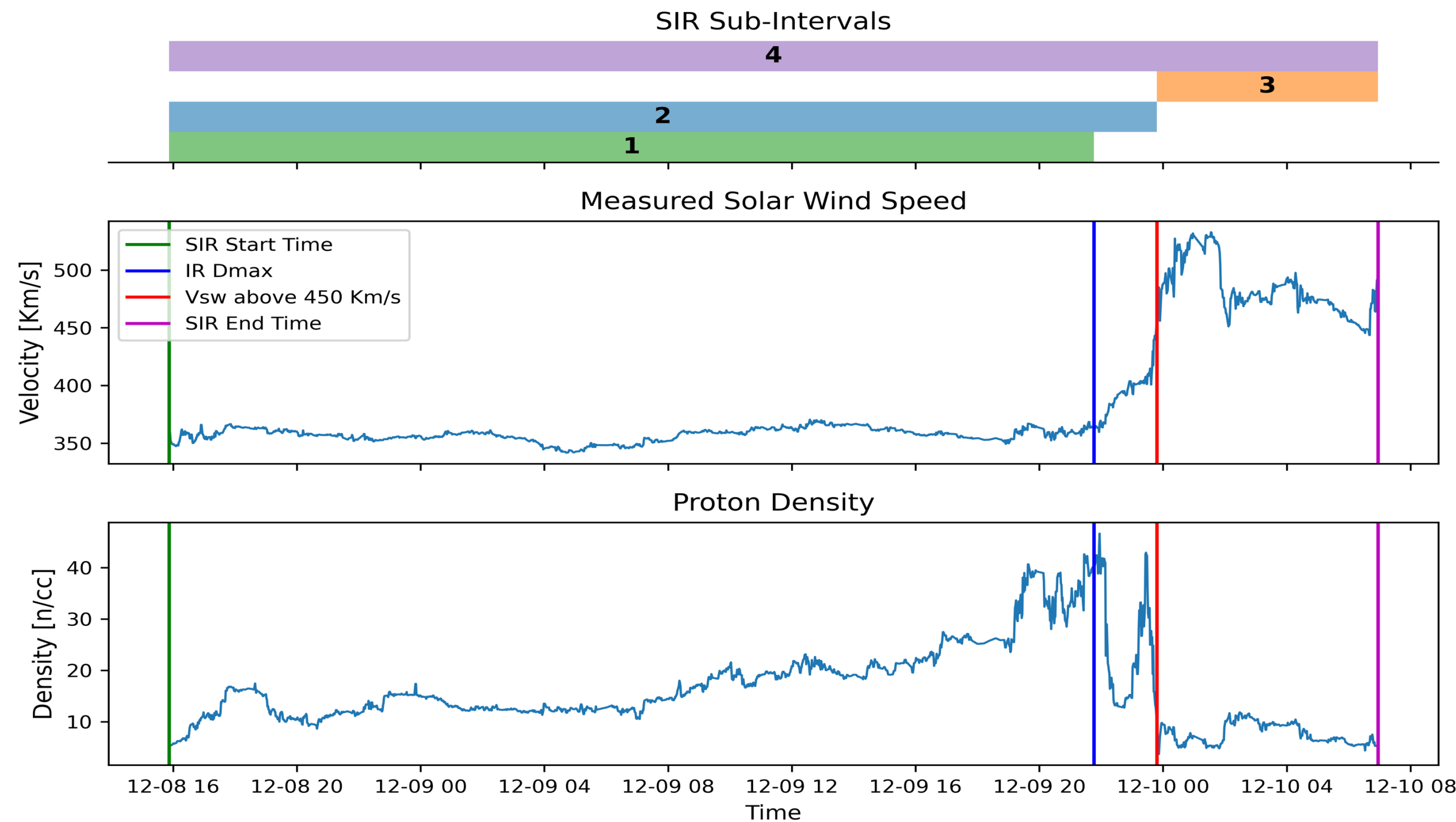
~ Methodology ~

We executed this analysis by following these steps:

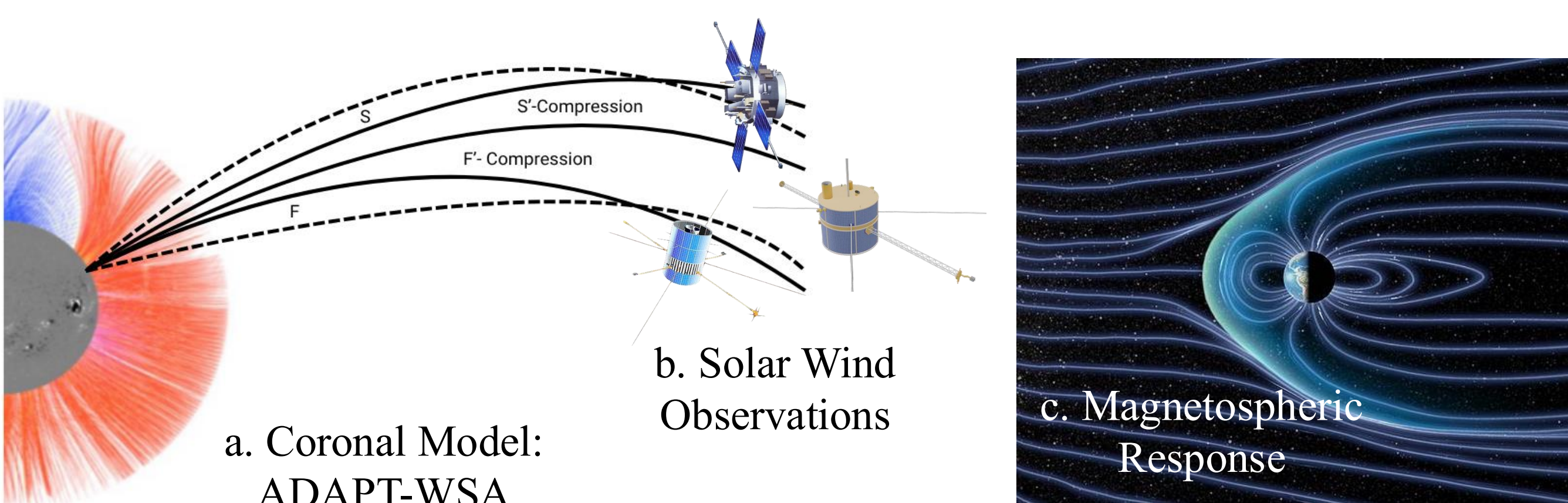
- **Event list:** 208 total SIR events, 149 with derived source-region parameters via ADAPT-WSA.
- **Calculated the Akasofu Parameter¹:**
 - **Definition:** A solar wind coupling function that combines solar wind speed, magnetic field strength, and IMF orientation to estimate the rate of energy transferred into Earth's magnetosphere.
$$\varepsilon = \frac{4\pi}{\mu_0} V B^2 L_o^2 \sin^4\left(\frac{\theta}{2}\right)$$
- Integrate over time to estimate the total energy transferred into the magnetosphere.

~ Methodology ~

• Divided Our List into a Four-Interval Framework:



• Histograms of Coronal Model, Geospace, and Solar Wind Responses:



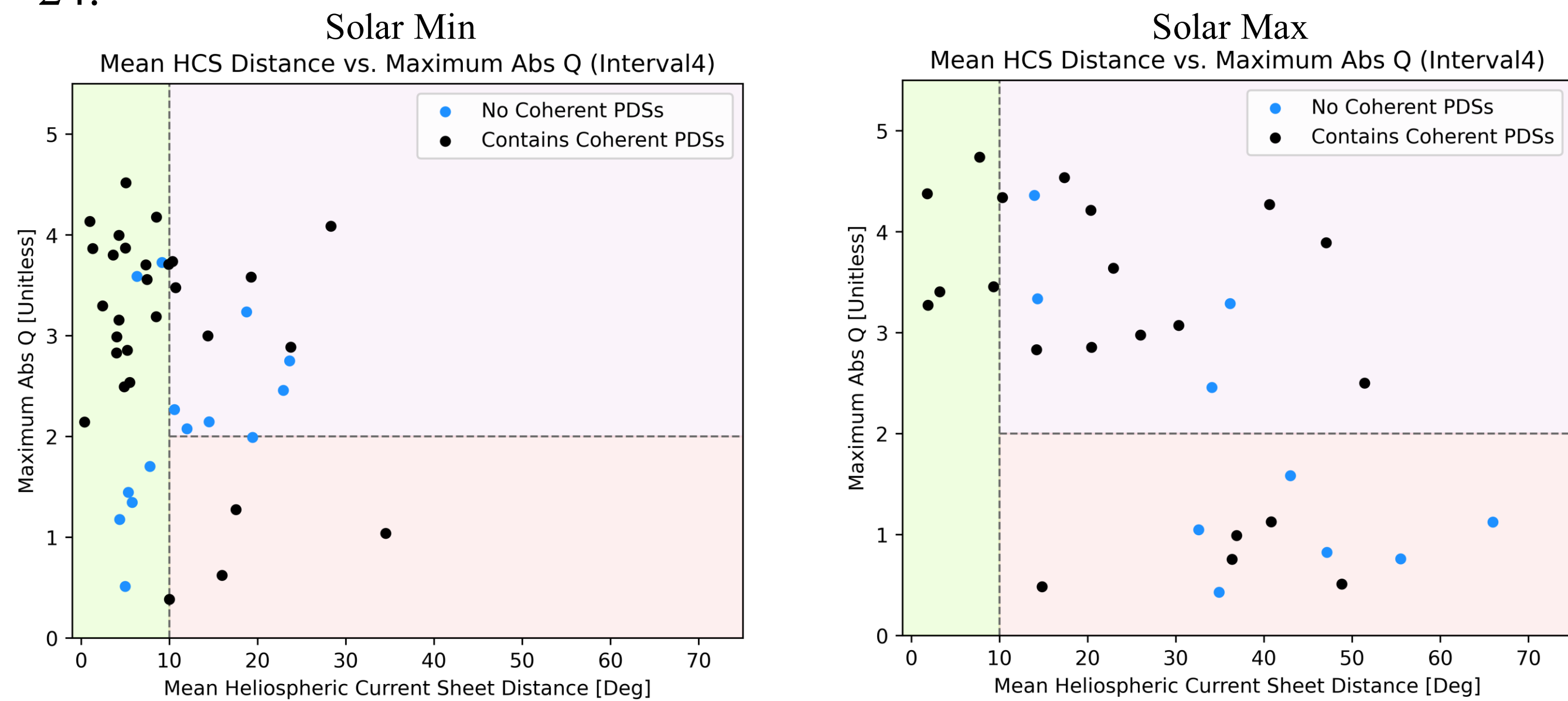
~ Conclusion & Future Work ~

Conclusion:

- Isolated SIRs with embedded PDSs drive moderate-to-strong geomagnetic activity, similar to canonical SIRs.
- Source-region modeling shows traces from S-web arcs and open/closed boundaries far from the heliospheric current sheet.

Future Work:

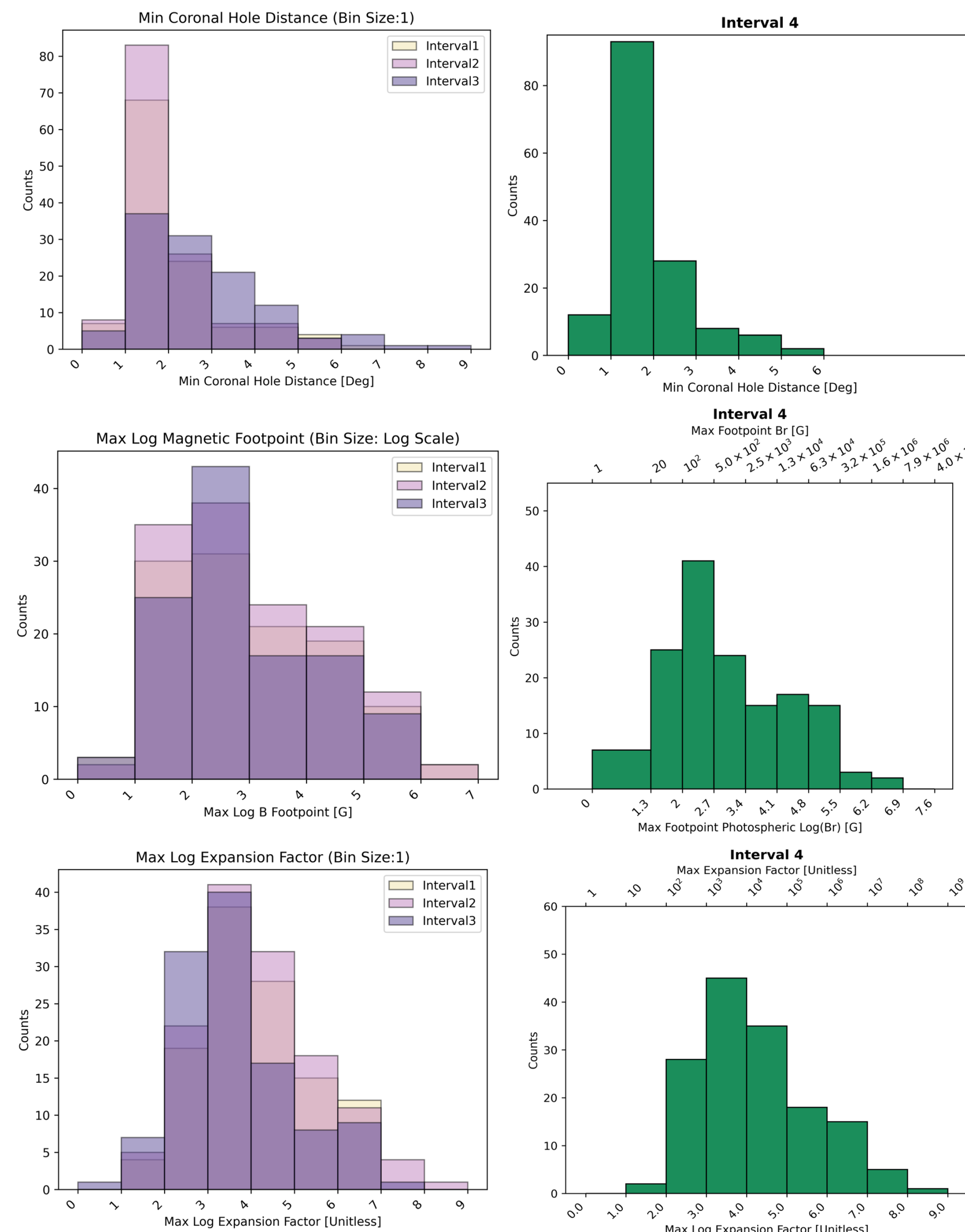
- Characterize PDS frequency distributions across SIR intervals back to their solar sources, and compare coherent vs. non-coherent PDS populations (normalized to total events) across the scatter plot quadrants.
- Compare results with solar maximum and minimum times for cycles 23 and 24.



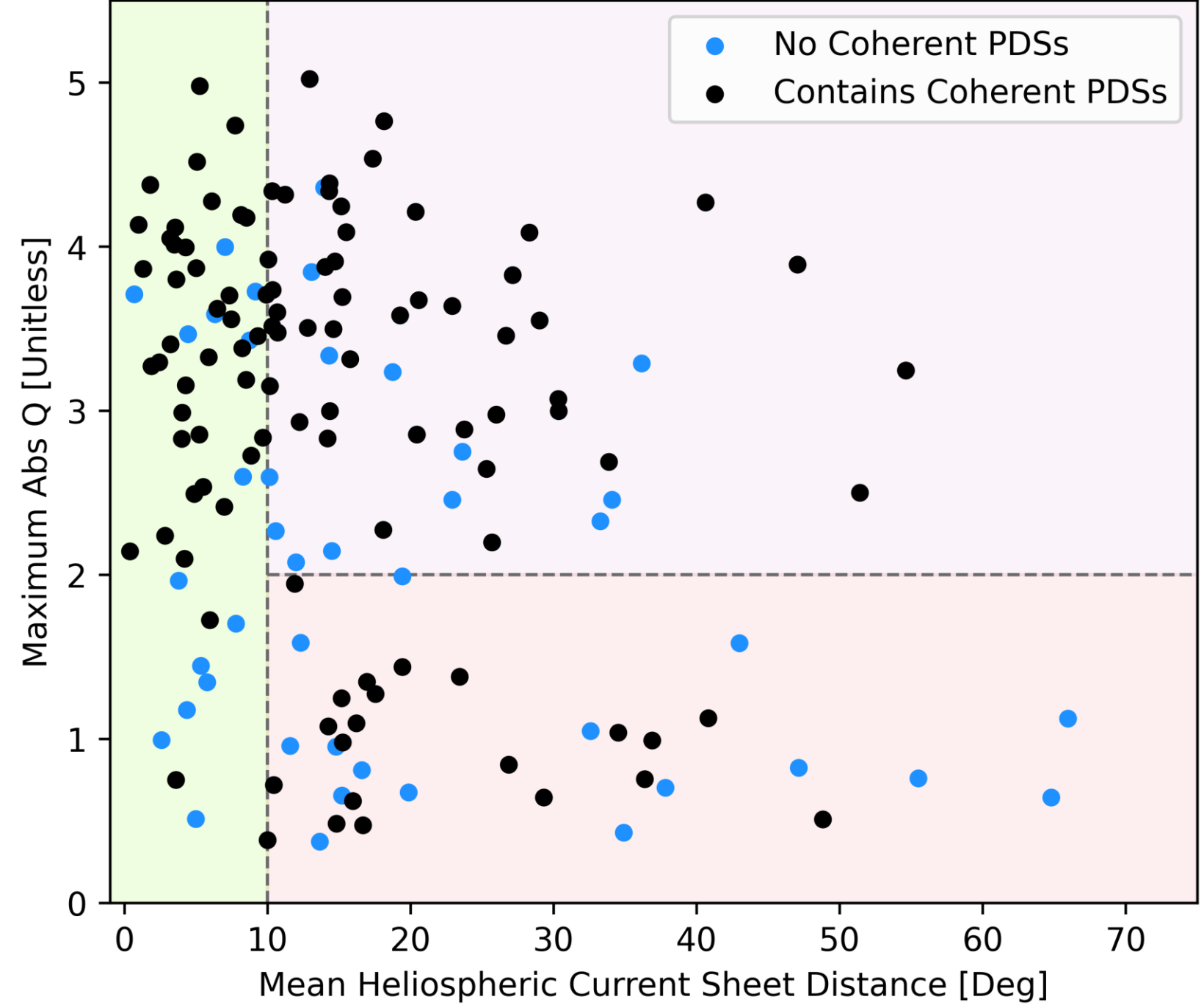
References:
1. Akasofu, S.-I. (1981). Energy coupling between the solar wind and the magnetosphere. Space Science Reviews, 28 (2), 121–190. doi:10.1007/BF00218810
2. Katsavrias, C., Di Matteo, S., Kepko, L., & Viall, N. M. (2025). The dependence of periodic density structures' amplitude and length scale on solar wind density within stream interaction regions. Astronomy & Astrophysics, 696 , L20. doi:10.1051/0004-6361/202554483

~ Results ~

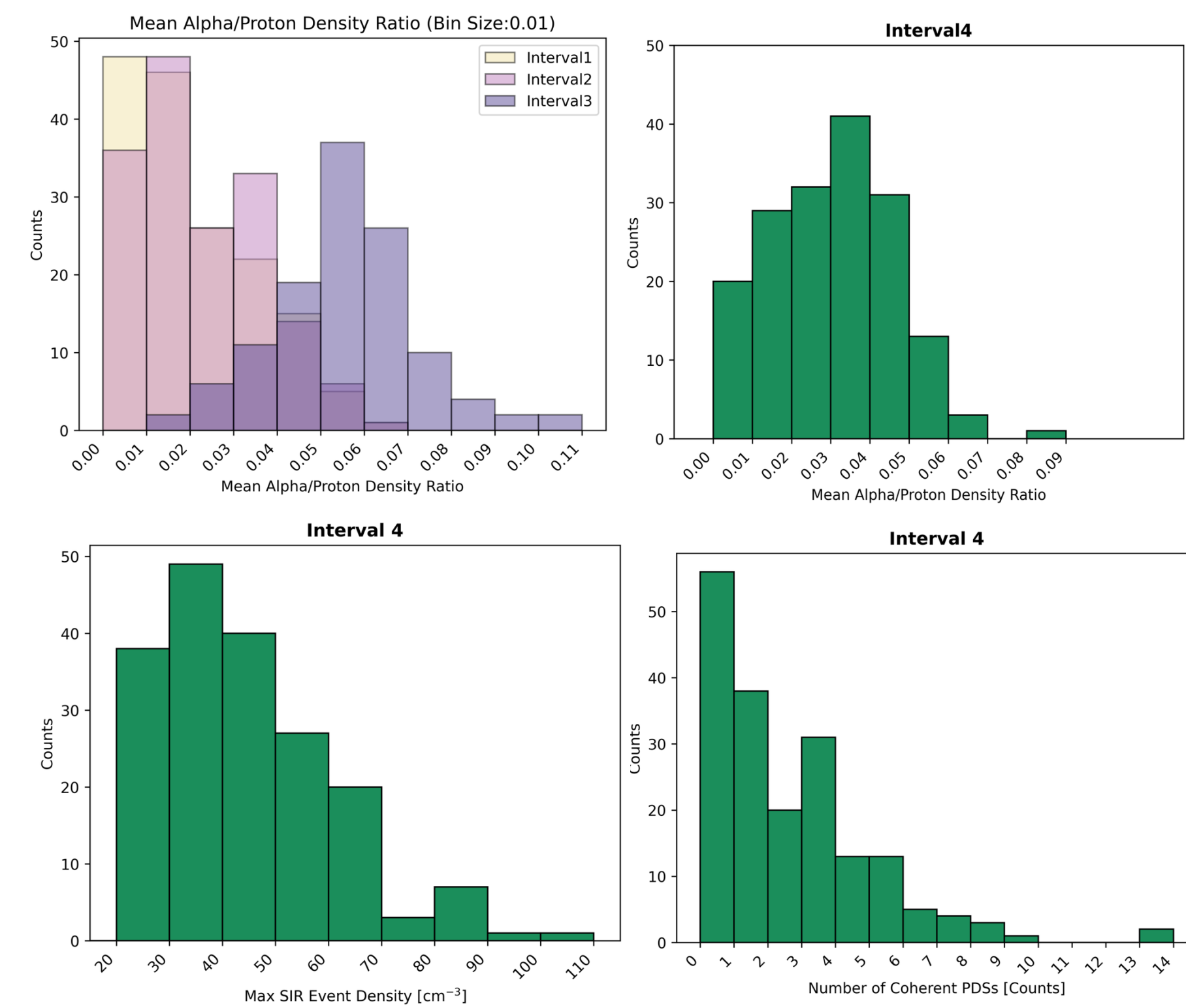
Source Origin Response



Mean HCS Distance vs. Maximum Abs Q (Interval 4)



Solar Wind Response



Geomagnetic Response

